

Ejercicios básicos con John the Ripper (Kali Linux)

Ejercicio 1: Crear hash SHA1 y ficheros de prueba

```
# hash SHA1 de "hello" es: aaf4c61ddcc5e8a2dabede0f3b482cd9aea9434d
printf 'aaf4c61ddcc5e8a2dabede0f3b482cd9aea9434d\n' > hashes_sha1.txt
printf 'hello\n' > wordlist_hello.txt

# comprobar contenido

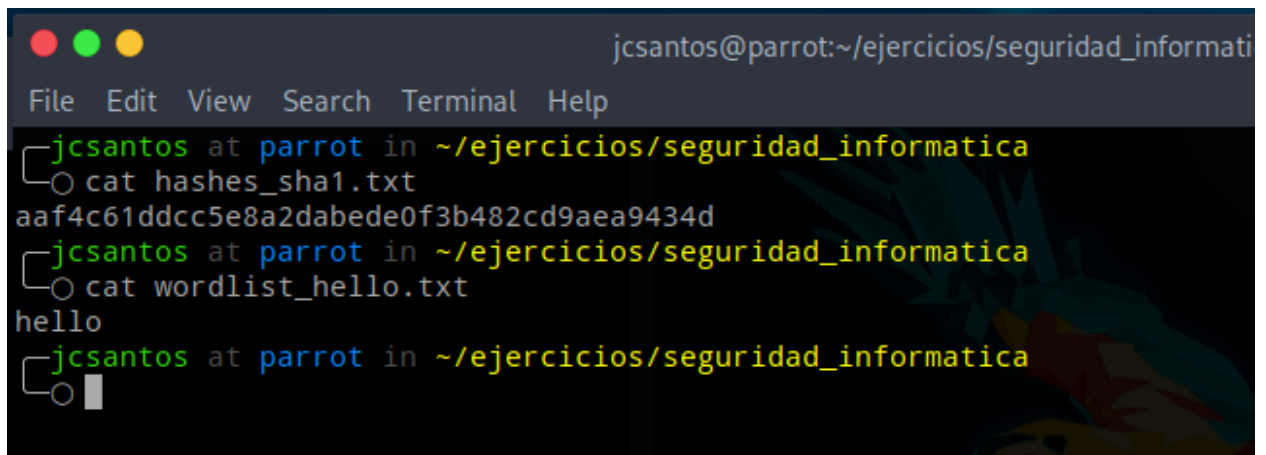
echo "HASHES:"

cat hashes_sha1.txt

echo

echo "WORDLIST:"

cat wordlist_hello.txt
```

A terminal window titled 'jcsantos@parrot:~/ejercicios/seguridad_informatica' with a menu bar (File, Edit, View, Search, Terminal, Help). The user 'jcsantos' is at the 'parrot' host in the directory '~/ejercicios/seguridad_informatica'. They execute 'cat hashes_sha1.txt' which outputs 'aaf4c61ddcc5e8a2dabede0f3b482cd9aea9434d'. Then they execute 'cat wordlist_hello.txt' which outputs 'hello'.

```
jcsantos@parrot:~/ejercicios/seguridad_informatica
File Edit View Search Terminal Help
jcsantos at parrot in ~/ejercicios/seguridad_informatica
└─○ cat hashes_sha1.txt
aaf4c61ddcc5e8a2dabede0f3b482cd9aea9434d
jcsantos at parrot in ~/ejercicios/seguridad_informatica
└─○ cat wordlist_hello.txt
hello
jcsantos at parrot in ~/ejercicios/seguridad_informatica
└─○
```

Intento con John (usa potfile temporal para no tocar ~/.john/john.pot)

```
john --pot=~/temp_sha1.pot --format=raw-sha1 --wordlist=wordlist_hello.txt
hashes_sha1.txt
```

A terminal window showing the execution of John the Ripper. The user runs 'sudo john --pot=~/temp_sha1.pot --format=raw-sha1 --wordlist=wordlist_hello.txt hashes_sha1.txt'. The output shows it loaded 1 password hash, issued a warning about OpenMP support, and then successfully cracked the password 'hello' after 1g 0:00:00:00. The terminal also shows the completion of the session.

```
jcsantos@parrot:~/ejercicios/seguridad_informatica
File Edit View Search Terminal Help
jcsantos at parrot in ~/ejercicios/seguridad_informatica
└─○ sudo john --pot=~/temp_sha1.pot --format=raw-sha1 --wordlist=wordlist_hello.txt hashes_sha1.txt
Using default input encoding: UTF-8
Loaded 1 password hash (Raw-SHA1 [SHA1 256/256 AVX2 8x])
Warning: no OpenMP support for this hash type, consider --fork=4
Press 'q' or Ctrl-C to abort, almost any other key for status
Warning: Only 1 candidate left, minimum 8 needed for performance.
hello (??)
1g 0:00:00:00 DONE (2026-01-26 11:51) 100.0g/s 100.0p/s 100.0c/s 100.0C/s hello
Use the "--show --format=Raw-SHA1" options to display all of the cracked passwords reliably
Session completed.
jcsantos at parrot in ~/ejercicios/seguridad_informatica
└─○
```

Ejercicio 2: Crear ficheros de prueba

```
# hash SHA256 de "secret123"
```

```
printf 'fcf730b6d95236ecd3c9fc2d92d7b6b2bb061514961aec041d6c7a7192f592e4\n' >  
hashes_sha256.txt
```

```
# wordlist con la contraseña en claro
```

```
printf 'secret123\n' > wordlist_secret.txt
```

```
# comprobar contenido
```

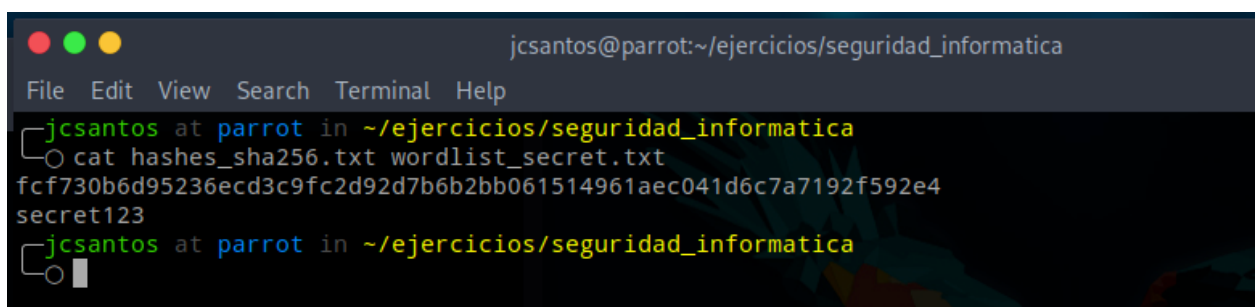
```
echo "HASH:"
```

```
cat hashes_sha256.txt
```

```
echo
```

```
echo "WORDLIST:"
```

```
cat wordlist_secret.txt
```

A terminal window with a dark background and light-colored text. The window title is 'jcsantos@parrot:~/ejercicios/seguridad_informatica'. The menu bar shows 'File', 'Edit', 'View', 'Search', 'Terminal', and 'Help'. The user 'jcsantos' is at the 'parrot' machine in the directory '~/ejercicios/seguridad_informatica'. They run the command 'cat hashes_sha256.txt wordlist_secret.txt'. The output shows the SHA256 hash 'fcf730b6d95236ecd3c9fc2d92d7b6b2bb061514961aec041d6c7a7192f592e4' followed by the password 'secret123' on a new line.

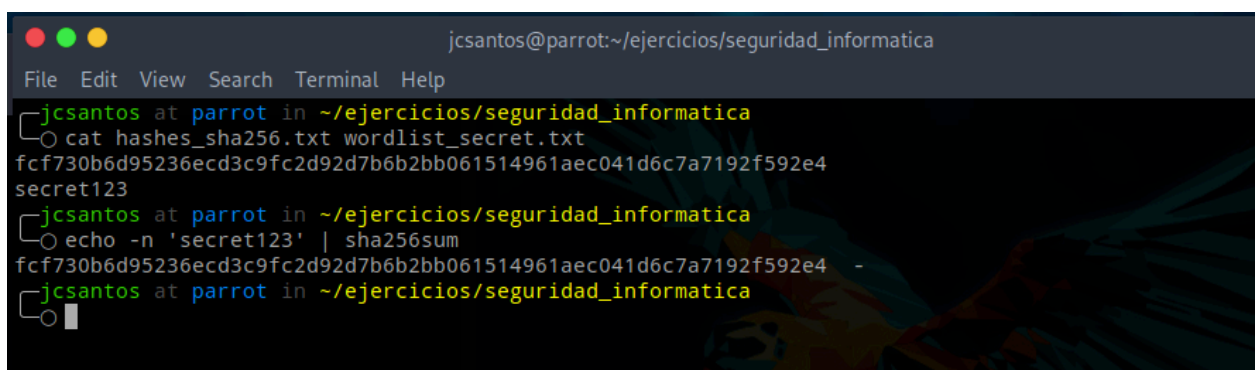
```
jcsantos@parrot:~/ejercicios/seguridad_informatica  
File Edit View Search Terminal Help  
jcsantos at parrot in ~/ejercicios/seguridad_informatica  
└─○ cat hashes_sha256.txt wordlist_secret.txt  
fcf730b6d95236ecd3c9fc2d92d7b6b2bb061514961aec041d6c7a7192f592e4  
secret123  
jcsantos at parrot in ~/ejercicios/seguridad_informatica  
└─○
```

2. Verificar localmente el hash (opcional)

```
echo -n 'secret123' | sha256sum
```

```
# salida esperada:
```

```
fcf730b6d95236ecd3c9fc2d92d7b6b2bb061514961aec041d6c7a7192f592e4 -
```

A terminal window showing the verification of the hash. The user runs 'echo -n 'secret123' | sha256sum'. The output is 'fcf730b6d95236ecd3c9fc2d92d7b6b2bb061514961aec041d6c7a7192f592e4 -', which matches the hash in the first file.

```
jcsantos@parrot:~/ejercicios/seguridad_informatica  
File Edit View Search Terminal Help  
jcsantos at parrot in ~/ejercicios/seguridad_informatica  
└─○ cat hashes_sha256.txt wordlist_secret.txt  
fcf730b6d95236ecd3c9fc2d92d7b6b2bb061514961aec041d6c7a7192f592e4  
secret123  
jcsantos at parrot in ~/ejercicios/seguridad_informatica  
└─○ echo -n 'secret123' | sha256sum  
fcf730b6d95236ecd3c9fc2d92d7b6b2bb061514961aec041d6c7a7192f592e4 -  
jcsantos at parrot in ~/ejercicios/seguridad_informatica  
└─○
```

3. Ejecutar John (potfile temporal para no tocar ~/.john/john.pot)

```
john --pot=./temp_sha256.pot --format=raw-sha256 --wordlist=wordlist_secret.txt hashes_sha256.txt
```

```
jcsantos at parrot in ~/ejercicios/seguridad_informatica
○ sudo john --pot=./temp_sha256.pot --format=raw-sha256 --wordlist=wordlist_secret.txt hashes_sha256.txt
Using default input encoding: UTF-8
Loaded 1 password hash (Raw-SHA256 [SHA256 256/256 AVX2 8x])
Warning: poor OpenMP scalability for this hash type, consider --fork=4
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
Warning: Only 1 candidate left, minimum 32 needed for performance.
secret123 (?)
1g 0:00:00:00 DONE (2026-01-26 11:54) 100.0g/s 100.0p/s 100.0c/s 100.0C/s secret123
Use the "--show --format=Raw-SHA256" options to display all of the cracked passwords reliably
Session completed.
jcsantos at parrot in ~/ejercicios/seguridad_informatica
○
```

Ejercicio 3: Ejemplo SHA-256

Perfecto — otro ejemplo SHA-256, paso a paso. Usaré la contraseña MiClave!2025. Su SHA-256 es:

c0f2bd9f0e58d2395799ee0b4802c8a1e28c8a4665af33e785c5cd4d255d75be

Copia/pega estos comandos en tu terminal (Kali/Parrot):

1. Crear ficheros de prueba

```
printf 'c0f2bd9f0e58d2395799ee0b4802c8a1e28c8a4665af33e785c5cd4d255d75be\n'
> hashes_sha256_b.txt
```

```
printf 'MiClave!2025\n' > wordlist_miclave.txt
```

2. Verificar localmente el hash (opcional)

```
echo -n 'MiClave!2025' | sha256sum
```

salida esperada:

```
# c0f2bd9f0e58d2395799ee0b4802c8a1e28c8a4665af33e785c5cd4d255d75be -
```

3. Probar con John (potfile temporal para no tocar ~/.john/john.pot)

```
john --pot=./temp_sha256_b.pot --format=raw-sha256 --wordlist=wordlist_miclave.txt  
hashes_sha256_b.txt
```

```
jcsantos at parrot in ~/ejercicios/seguridad_informatica  
└─○ sudo john --pot=./temp_sha256_b.pot --format=raw-sha256 --wordlist=wordlist_miclave.txt hashes_s  
ha256_b.txt  
Using default input encoding: UTF-8  
Loaded 1 password hash (Raw-SHA256 [SHA256 256/256 AVX2 8x])  
Warning: poor OpenMP scalability for this hash type, consider --fork=4  
Will run 4 OpenMP threads  
Press 'q' or Ctrl-C to abort, almost any other key for status  
Warning: Only 1 candidate left, minimum 32 needed for performance.  
MiClave!2025 (?)  
1g 0:00:00:00 DONE (2026-01-26 11:56) 100.0g/s 100.0p/s 100.0c/s 100.0C/s MiClave!2025  
Use the "--show --format=Raw-SHA256" options to display all of the cracked passwords reliably  
Session completed.  
jcsantos at parrot in ~/ejercicios/seguridad_informatica  
└─○
```

```
jcsantos@parrot:~/ejercicios/seguridad_informatica  
File Edit View Search Terminal Help  
jcsantos at parrot in ~/ejercicios/seguridad_informatica  
└─○ echo -n 'Que_pasa_Bro_2025' | sha256sum  
3c93fdd8e478f4c3f7334db2f4648d9733b774b7b46ed5178ac812f01d5ef079 -  
jcsantos at parrot in ~/ejercicios/seguridad_informatica  
└─○ printf '3c93fdd8e478f4c3f7334db2f4648d9733b774b7b46ed5178ac812f01d5ef079\n' > hash_256_JC.txt  
jcsantos at parrot in ~/ejercicios/seguridad_informatica  
└─○ printf 'Que_pasa_Bro_2025' >> wordlist_miclave.txt  
jcsantos at parrot in ~/ejercicios/seguridad_informatica  
└─○ sudo john --pot=./temp_sha256_b.pot --format=raw-sha256 --wordlist=wordlist_miclave.txt hash_256  
_JC.txt  
Using default input encoding: UTF-8  
Loaded 1 password hash (Raw-SHA256 [SHA256 256/256 AVX2 8x])  
Warning: poor OpenMP scalability for this hash type, consider --fork=4  
Will run 4 OpenMP threads  
Press 'q' or Ctrl-C to abort, almost any other key for status  
Warning: Only 2 candidates left, minimum 32 needed for performance.  
Que_pasa_Bro_2025 (?)  
1g 0:00:00:00 DONE (2026-01-26 12:00) 100.0g/s 200.0p/s 200.0c/s 200.0C/s MiClave!2025..Que_pasa_Bro  
_2025  
Use the "--show --format=Raw-SHA256" options to display all of the cracked passwords reliably  
Session completed.  
jcsantos at parrot in ~/ejercicios/seguridad_informatica  
└─○
```